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A note on sequential auctions with multi-unit demand

Zhiyong Yao*, Zhiguo Xiao

School of Management, Fudan University, 670 Guoshun Road, 200433, China



HIGHLIGHTS

- We analyze sequential first-price auctions with multi-unit demand.
- We identify the “extraction effect”.
- We compare the revenues for the sequential and the simultaneous auctions.

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ABSTRACT

Within the independent private value paradigm, this note first analyzes two-round sequential first-price auctions with multi-unit demand. We show that the expected price in the first round is strictly lower than that in the second round due to the “extraction effect”. We then compare the revenues for the sequential auctions and the simultaneous auctions. We show that the discriminatory auction, the Vickrey auction, and the sequential second-price auctions generate the same revenue for the seller, followed in order by the sequential first-price auctions, and by the uniform-price auction.

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1. Introduction

Multi-unit demand is common in real-world sequential auctions. For example, a few large corporations compete for government defense or construction contracts year after year using first-price sealed bid auctions (FPA henceforth). The same retailers regularly attend the Dutch auctions for nondurable wholesale goods, such as flowers or fish. In the B2B context, firms participate in sequences of auctions to procure inputs again and again. In the B2C context, consumers have multi-unit demand over time for many nondurable goods, because today's buyers will not remain satiated in the future. Yet most sequential auction literature focuses on single-unit demand (Maskin and Riley, 1989; McAfee and Vincent, 1993; Engelbrecht-Wiggans, 1994; Milgrom and Weber, 2000, and many others). There are a few papers that focus on sequential auctions with multi-unit demand, but very often their attentions are restricted to the second-price auctions (SPA henceforth) (Katzman, 1999; Liu, 2011).

This article first analyzes the sequential FPA with multi-unit demand using a simple model; then we compare its revenue with the sequential SPA and simultaneous auctions. In our independent private value (IPV henceforth) environment, two identical goods will be sold to two bidders by a sequence of two-round FPA. Both bidders wish to win both goods. They draw their values for the first

item from two possible valuations,¹ and the value for the second item is either constant or diminishing compared to the first. We assume that only the winning bid is announced after the first round.² The bidders will use this information to update their beliefs about the valuation of their opponents for the second round. Once a particular bidder has won, her behavior and valuation are different from the other. In other words, even though the bidders are symmetric ex ante, asymmetries arise after the first round. This characteristic contrasts with the single-unit demand sequential auctions, where each round remains symmetric since the winner drops out.

After characterizing the equilibrium for the sequential FPA, we show that the expected price in the second round is strictly higher than that of the first. The reason that the buyers bid lower in the first round is due to the “extraction effect”, which consists of the “surplus extraction” and the “information extraction”. On the one hand, the high-type player's expected payoff in the second round is decreasing in the first-round winning bid. Therefore, she has the incentive to shade her bid in the first round to induce less aggressive bid in the second round. This is the surplus extraction. On the other hand, even if she loses by bidding low in the first round, she then knows the winner's valuation but disguises her own since only the winning bid is announced. This is the information extraction. It is

¹ Note that we do not consider continuous valuations here because there does not exist a monotonic equilibrium in the continuous type case when the winning bid is disclosed (Enkhbayar, 2004).

² This assumption is in accordance with government procurement statutes and with practices in many auctions (McAfee and Vincent, 1997).

* Corresponding author.

E-mail addresses: zyy@fudan.edu.cn (Z. Yao), zhiguo_xiao@fudan.edu.cn (Z. Xiao).

the extraction effect which makes the buyers bid lower in the first round. This result is different from the single-unit demand literature, where the price in the second auction is, on average, the same as the price in the first auction (Weber, 1983; Jeitschko, 1998).

We then compare the sequential FPA and the sequential SPA. We show that sequential SPA dominates the sequential FPA in terms of revenue for the seller. The intuition is as follows. For the sequential SPA, it is the weakly dominant strategy to bid one's true value for the second round, and in the first round the bidders will bid their values for the second item. However, the bidders in the sequential FPA bid lower for the first item as we have explained above, which also leads to less aggressive bids, on average, for the second item. We also compare the revenues between the sequential auctions and the simultaneous auctions. We show that the discriminatory auction, the Vickrey auction, and the sequential SPA generate the same revenue for the seller, followed in order by the sequential FPA, and by the uniform-price auction. This result of revenue comparison is different from the literature with single-unit demand and IPV environment, where the celebrated "revenue equivalence theorem" (Vickrey, 1961; Myerson, 1981; Riley and Samuelson, 1981) holds no matter whether the units are sold sequentially or simultaneously (Weber, 1983; Maskin and Riley, 1989; Jeitschko, 1998).

Ortega-Reichert (1968) is the first one to study two-bidder, two-round, FPA with multi-unit demand in a dependent private value setting, in which each bidder draws his or her type before each round starts and all bids are announced after the first round.³ In a similar setting, Hausch (1986) identifies the "information effect" and the "deception effect" in a two-signal model, and shows that either effect may dominate the other, leading the seller sometimes to prefer sequential auctions and sometimes to prefer simultaneous auctions. Hausch (1988) generalizes the model of Ortega-Reichert (1968) by studying both sequential FPA and SPA. This article is different from Ortega-Reichert (1968) and Hausch (1988) in the following ways: the bidders draw their values only once, which is at the beginning of the first round; and only the winning bid is announced after the first round, which makes the second auction an asymmetric one. However, Ortega-Reichert (1968) and Hausch (1988) are still symmetric auctions, and they are able to characterize pure strategy symmetric equilibrium. Therefore, although they identify the so-called "deception effect", in equilibrium there is no deception. In our model, we identify the "extraction effect" as described above.

Weber (1983) discusses the complexity that arises when bidders desire multiple units. Milgrom and Weber (2000) consider a wide class of sequential auctions and show that with affiliation or common-value elements, the price path drifts upwards. In our article, the price path also drifts upwards, but our setting is IPV with multi-unit demand. Jeitschko (1998) also studies two-round, FPA with two-type bidders. He clearly identifies the potential impact of information transmission and learning when goods are auctioned sequentially. However, Jeitschko (1998) focuses on the single-unit demand. Katzman (1999) analyzes sequential SPA with multi-unit demands within the IPV paradigm. As mentioned before, the bidder has a weakly dominant strategy to bid her true value in the second round despite the information transmission and learning,

which simplifies the analysis. Liu (2011) constructs an ex post perfect equilibrium in a sequence of second-price auctions with two bidders where each bidder's marginal values for additional units are decreasing. The equilibrium implies an increasing price path and is ex post efficient. Caillaud and Mezzetti (2004) investigate sequential ascending auctions with multi-unit demand. They characterize the equilibrium reserve prices and study their properties. Fevrier (2003) studies two-round FPA in which only the identity of the winner will be announced after the first round. His numerical results show that the simultaneous sale of two goods is preferable to a sequential sale. In a procurement environment, Thomas (2010) compares different information disclosure policies in sequential FPA and suggests that buyers may pay less by revealing minimal or extensive information, rather than an intermediate amount.

In the next section we first describe the basic environment, then characterize the equilibrium for the sequential FPA. Revenue comparison is conducted in Section 3. Section 4 concludes.

2. Model

Two identical objects are sold sequentially by two rounds FPA to two risk-neutral bidders, A and B. Each bidder values v for the first item and αv for the second item, where $0 < \alpha \leq 1$. The valuation v is binary, either low or high. Without loss of generality, we normalize the low and the high values to be 0 and $\theta(>0)$, respectively.⁴ Both bidders draw their own valuations v_i before the first round starts, where $i = A, B$. Each bidder knows her own value, but she only knows her opponent is a low (high) type bidder with probability $\underline{p}$ ($\bar{p}$), where $\underline{p} + \bar{p} = 1$. The two bidders will compete for the items in both rounds. We assume there is no discounting, and both auctions have zero reserve prices.

In the first round, both buyers submit their bids simultaneously. Then the winning bid b_1^w will be publicly announced. Next, in the second round both buyers again submit their bids simultaneously, and the winning bid b_2^w will be announced. The winner's payoff is $v_1^w - b_1^w$ if she wins the first item, and $(1 + \alpha)v_1^w - b_1^w - b_2^w$ if she wins both rounds.

The equilibrium concept for the sequential auctions is the perfect Bayesian equilibrium (henceforth, PBE). It consists of a pair of bidding strategies $(\beta_{i1}(\cdot), \beta_{i2}(\cdot))$ and a belief r_i , which is the belief of bidder i about her opponent j 's type once the outcome of the first round is realized, such that: (i) For each possible realization of the first-round bids (b_{i1}, b_{j1}^w) , β_{i2} is optimal under her belief r_i given β_{j2} , (ii) β_{i1} is optimal given β_{j1} , β_{j2} , and (iii) The belief r_i is generated by using Bayes' rule whenever possible, where $i, j = A, B$; $i \neq j$.

As a useful benchmark, we first discuss the standard one-shot FPA. Suppose there is only one item sold in FPA. In equilibrium, the low-type buyer (with valuation 0) bids 0, and the high-type buyer (with valuation θ) implements a mix strategy. The high-type can win with probability $\underline{p}$ if she bids just above 0. Hence, she submits bid b to satisfy the condition $\underline{p}\theta = (\theta - b)(\underline{p} + \bar{p}H(b))$, where $H(b)$ is the probability that her bid exceeds her high-type opponent. Thus, the high-type bidder will submit bid $b \in [0, \bar{p}\theta]$ with c.d.f. $H(b) = \frac{\underline{p}}{\bar{p}(\theta - b)}$, and her expected payoff is $\underline{p}\theta$.

³ The game timing of Ortega-Reichert (1968) is the following: (1) nature secretly draws a state variable from a commonly-known distribution; (2) each player independently draws his type for the first item from a common distribution which is conditioned on the state variable; (3) each player bids for the first item after which both bids are revealed; (4) each player independently draws his type for the second item from the earlier distribution still conditioned on the state variable; and (5) each player bids on the second item. For more discussions on Ortega-Reichert (1968), please see Hausch (1986, 1988).

⁴ It is without loss of generality to assume that the low-type value is 0. Suppose instead that the low-type value is $\underline{\theta} > 0$. Notice first that a $\underline{\theta}$ bidder will never bid more than $\underline{\theta}$ because, if she did, she would win the auction with positive probability, inducing a negative expected payoff. Second, suppose that the $\underline{\theta}$ player bids $\underline{b} < \underline{\theta}$. The winning chance of the other bidder, either a high or a low type, increases discontinuously by bidding just more than $\underline{b}$, thereby raising his expected payoff. Therefore, it is not an equilibrium by the $\underline{\theta}$ type to bid smaller than $\underline{\theta}$. In summary, just like the 0-type bidder, the $\underline{\theta}$ bidder will bid her true value and her expected payoff is 0.

If there are two rounds, there is an information transmission between these two auctions. The bidders will update their beliefs after the first-round winning bid is publicly announced. Obviously, the beliefs for the first-round winner and the loser will be different. Thus, the second round becomes an asymmetric auction.

2.1. Equilibrium

In the sequential FPA, a low-type buyer always bids 0. We therefore focus our analysis on the high-type. Suppose that in the first round the high-type buyer i bids b_1 , her expected payoff can be written as $\pi_1(b_1) = (\theta - b_1)(\underline{p} + \bar{p}F_1(b_1))$, where $F_1(b_1)$ is the probability that her bid exceeds her high-type opponent and $\underline{p} + \bar{p}F_1(b_1)$ is the probability of being a winner of the first round.

After the winning bid in the first auction is publicly announced, the buyers update their beliefs before the second round. Let $\bar{r}$ be the posterior belief that the first round high-type winner (W henceforth) believes that the loser is also a high-type, then $\bar{r} = \frac{\bar{p}F_1(b_1)}{\underline{p} + \bar{p}F_1(b_1)}$. It is obvious that $\bar{r} \leq \bar{p}$. Thus, at the beginning of the second round, it is common knowledge that W has a valuation of $\alpha\theta$,⁵ and W knows that the loser has valuation θ and 0 with respective probabilities $\bar{r}$ and $\underline{r}$, where $\underline{r} = 1 - \bar{r}$.

Both W and the high-type loser (L henceforth) will implement mixed strategies in the second round. W will bid $b_2^w \in [0, \bar{b}_2^w]$ with c.d.f. F_2^w , and L will bid $b_2^l \in [0, \bar{b}_2^l]$ with c.d.f. F_2^l . In addition, it must be the case that $\bar{b}_2^w = \bar{b}_2^l = \bar{b}_2$ in equilibrium. To see why, consider what would happen if $\bar{b}_2^w > \bar{b}_2^l$. Then W could win the second round with probability 1 by bidding $\bar{b}_2^w$, and she could increase her payoff by bidding slightly less than $\bar{b}_2^w$.

In the second round, W can win with probability (at least) $\underline{r}$ if she bids (just above) 0. Her expected payoff from doing so is (at least) $\alpha\underline{r}\theta$. Therefore, bidder W submits b_2^w such that:

$$\alpha\underline{r}\theta = [\underline{r} + \bar{r}F_2^l(b_2^w)](\alpha\theta - b_2^w);$$

bidder L submits b_2^l such that:

$$\theta - \bar{b}_2 = (\theta - b_2^l)F_2^w(b_2^l).$$

Therefore, we obtain:

$$\bar{b}_2 = \frac{\alpha\theta\bar{p}F_1(b_1)}{\underline{p} + \bar{p}F_1(b_1)}, \quad F_2^l(b_2) = \frac{pb_2}{(\alpha\theta - b_2)\bar{p}F_1(b_1)},$$

$$F_2^w(b_2) = \frac{\theta[\underline{p} + (1 - \alpha)\bar{p}F_1(b_1)]}{(\theta - b_2)(\underline{p} + \bar{p}F_1(b_1))}.$$

It is easy to show that $F_2^l(b_2)$ first order stochastically dominates $F_2^w(b_2)$, which means that L bids more aggressively than W .⁶

If the high-type wins the first round, her second-round payoff is $\alpha\underline{r}\theta$ as we have shown. However, if the high-type bidder i loses the first round, her second-round expected payoff is:

$$\begin{aligned} u_2(b_1) &= E(\theta(1 - \alpha\bar{r}) \mid b_1^w \geq b_1) \\ &= \int_{b_1}^{\bar{b}_1} \left\{ \theta - \alpha\theta \frac{\bar{p}F_1(b_1^w)}{\underline{p} + \bar{p}F_1(b_1^w)} \right\} f_1(b_1^w) db_1^w \\ &= (1 - \alpha)\theta(1 - F_1(b_1)) - \alpha\theta \frac{\underline{p}}{\bar{p}} \ln(\underline{p} + \bar{p}F_1(b_1)), \end{aligned}$$

⁵ Note that the high-type player will not bid zero in the first round. By bidding infinitesimally more than zero, the bidder's chance of winning increases discontinuously, while her payment scarcely rises if she wins. Therefore, her expected payoff increases.

⁶ Note that $F_2^w(0) > 0$, and it actually means that W will bid infinitesimally more than zero with positive probability. The high-type player will not bid exactly zero, the reason is the same as that of footnote 5. For more details, please see Maskin and Riley (1985, pp. 153–154).

where $f_1(b)$ is the p.d.f. of the bids in the first round. Then the second-round expected payoff for the high-type is:

$$\pi_2(b_1) = (\underline{p} + \bar{p}F_1(b_1))\alpha\underline{r}\theta + \bar{p}(1 - F_1(b_1))u_2(b_1),$$

where $\underline{p} + \bar{p}F_1(b_1)$ is the probability of being the winner of the first round, and $\bar{p}(1 - F_1(b_1))$ is the probability of being the first-round loser. Therefore, bidder i 's total expected payoff for the whole game is:

$$\pi(b_1) = (\theta - b_1 + \alpha\underline{r}\theta)(\underline{p} + \bar{p}F_1(b_1)) + \bar{p}(1 - F_1(b_1))u_2(b_1). \quad (1)$$

She implements a mixed strategy in the first round. When $b_1 = \bar{b}_1$, (1) becomes:

$$\pi(\bar{b}_1) = \theta - \bar{b}_1 + \alpha\underline{p}\theta. \quad (2)$$

When $b_1 = \underline{b}_1 = 0$, (1) becomes:

$$\pi(\underline{b}_1) = \theta[1 + \alpha(\underline{p} - \bar{p} - \underline{p} \ln \underline{p})]. \quad (3)$$

Letting (2) equal (3), we obtain:

$$\bar{b}_1^* = \alpha\theta(\underline{p} \ln \underline{p} + \bar{p}).$$

Thus, the high-type bidder i will submit a bid $b_1^* \in [0, \bar{b}_1^*]$ with c.d.f. $F_1^* \equiv F_1(b_1^*)$, which is characterized by $\pi(b_1) = \pi(\underline{b}_1)$ or:

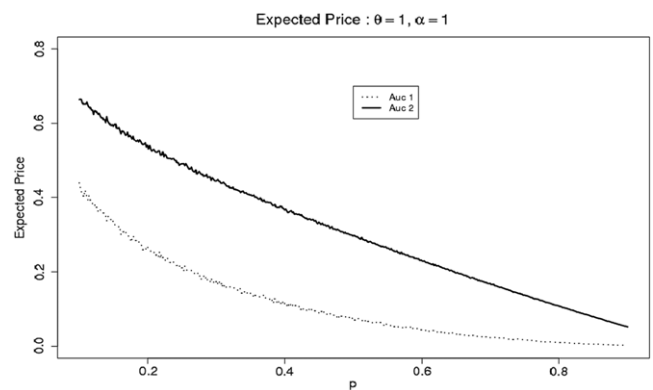
$$\begin{aligned} (\theta - b_1^*)(\underline{p} + \bar{p}F_1^*) + \theta(1 - F_1^*)[(1 - F_1^*)(1 - \alpha)\bar{p} \\ - \alpha\underline{p} \ln(\underline{p} + \bar{p}F_1^*)] = \theta(1 - \alpha\underline{p} \ln \underline{p} - \alpha\bar{p}). \end{aligned}$$

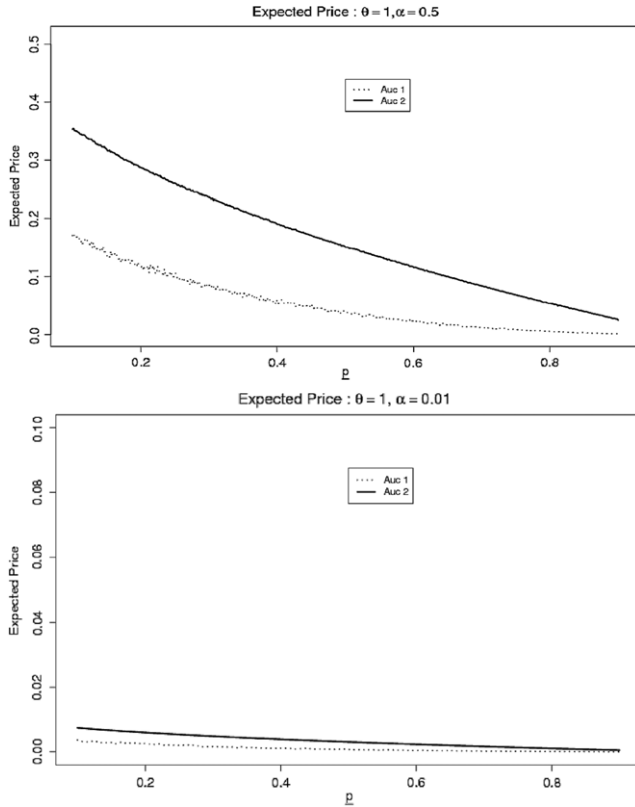
And the expected payoff for the high-type player is $\theta[1 + \alpha(\underline{p} - \bar{p} - \underline{p} \ln \underline{p})]$.

In summary of the above results, we obtain the following proposition:

Proposition 1. In the sequential FPA, there exists a PBE in which the high-type buyer randomizes over $[0, \alpha\theta(\underline{p} \ln \underline{p} + \bar{p})]$ with c.d.f. F_1^* in the first round, and in the second round the first-round high-type winner (loser) randomizes over $[0, \frac{\alpha\theta\bar{p}F_1^*(b_1^w)}{\underline{p} + \bar{p}F_1^*(b_1^w)}]$ with c.d.f. $F_2^{w*} = \frac{\theta[\underline{p} + (1 - \alpha)\bar{p}F_1^*(b_1^w)]}{(\theta - b_2^w)(\underline{p} + \bar{p}F_1^*(b_1^w))}$ ($F_2^{l*} = \frac{pb_2}{(\alpha\theta - b_2)\bar{p}F_1^*(b_1^w)}$). And the high-type's expected payoff is $\theta[1 + \alpha(\underline{p} - \bar{p} - \underline{p} \ln \underline{p})]$.

It is difficult to compare the bids in the two rounds directly. Therefore, we simulate the expected prices given specific parameter values. In each of the following graphs, the horizontal axis represents $\underline{p}$ and the vertical axis represents the expected price. In the first figure, $\theta = 1$ and $\alpha = 1$, and in the second, θ remains 1, but α drops to 0.5.





In these figures, we can see the expected prices in both rounds decrease as p increases. With other things unchanged, the bidders bid less in both rounds as α decreases. These results are quite intuitive. Our simulation results also show that the expected price in the first round is strictly lower than that of the second. The same pattern holds even when α approaches zero. For example, let $\alpha = 0.01$, and the figure shows that the pattern of an increasing price path persists.

It is known that in sequential auctions with single-unit demand, the expected sale prices in each auction are identical within the IPV framework; with affiliated values, the expected prices rise in later auctions because early auctions release information, thereby reducing winner's curse concerns (Weber, 1983). But with multi-unit demand, the outcomes are different. In our multi-unit demand case, the buyers bid lower in the first auction than in the second auction despite the IPV environment. The reason is due to the "extraction effect". The extraction effect has two parts. The first part is the surplus extraction. Besides the first-round payoff, the winning bid b_1^w has impact on the expected payoff of the second round. As indicated above, the second-round distribution F_2^w and F_2^L are supported on $[0, \bar{b}_2(b_1^w)]$, where $\bar{b}_2(b_1^w) = \alpha \theta \bar{r}(b_1^w)$ is increasing in b_1^w . The winner of the first round has an expected second-round payoff $\alpha \theta - \bar{b}_2(b_1^w)$, since she could bid $\bar{b}_2(b_1^w)$ to win for sure. Note that this expected payoff is decreasing in b_1^w . Thus, the high-type bidder has the incentive to shade her bid in the first round to induce less aggressive bidding in the second round. The second part is the information extraction. When a high-type player bids low and she loses, she now knows the winner's valuation but disguises her own since only the winning bid is announced. It is the extraction effect which makes the high-type player bids lower in the first round than in the second round.

3. Revenue comparison

In this section, we first analyze the sequential SPA and compare its revenue with the sequential FPA. Then we discuss the simultaneous auctions and compare the revenues between the sequential auctions and the simultaneous auctions.

3.1. Sequential second price auctions

Suppose instead that the two items are auctioned out by two-round SPA; we have the following lemma:

Lemma 1. *In the sequential SPA, there exists a PBE in which the high-type bids $\alpha\theta$ in the first round. If she wins, she bids $\alpha\theta$; if she loses, she bids θ in the second round. And the high-type's expected payoff is $[1 + \alpha(p - \bar{p})]\theta$.*

Proof. By following the equilibrium strategy, the high-type player, say bidder A, will win both items at price 0 if B is a low type. She will win only one item at price $\alpha\theta$ and obtain payoff $\theta - \alpha\theta$ if facing a high-type opponent.

Clearly, in the second round, the buyers will bid their true values as there is only one round SPA left. Next, we show that a high-type bidder has no profitable deviation from bidding $\alpha\theta$ in the first round. Suppose that the high-type bidder A bids $b > \alpha\theta$ in the first round. If her opponent is a low type, she will win anyway at price 0. If her opponent is also a high-type and follows the prescribed strategy to bid $\alpha\theta$, bidder A will win the first round and pay $\alpha\theta$, and her payoff is $\theta - \alpha\theta$. Then bidder B will win the second round. Clearly, there is no gain to bid $b > \alpha\theta$. Suppose that player A bids $b < \alpha\theta$ in the first round. Again, if her opponent is a low type, she will win at price 0. If bidder B is a high type and bids $\alpha\theta$, bidder B wins the first round. But bidder A will win the second round and gets the payoff $\theta - \alpha\theta$. Again this is not a profitable deviation. It is easy to verify that the high-type's expected payoff is $[1 + \alpha(p - \bar{p})]\theta$. $\square$

It is obvious that, compared to the sequential SPA, the high-type obtains more in the sequential FPA. Given the total surpluses are the same for both auctions, the auctioneer necessarily receives more in the sequential SPA. The intuition for this result is as follows. For the sequential SPA, in the first round the players bid their values for the second item, as Lemma 1 indicates. In the second round, the players bid their true values, and the extra information (the revealed winning bid) obtained after the first round does not play a role here. However, the players in the sequential FPA bid lower for the first item due to the extraction effect as we have explained above, and this also leads to on average less aggressive bids for the second item.

Note that the sequential SPA is ex post efficient, but the sequential FPA is not. For example, suppose that both bidders are of the high type. In the sequential SPA, each of them obtains one object. However, in the sequential FPA it is possible that one of the bidders gets both objects.

3.2. Simultaneous auctions

So far we have focused only on sequential auctions. There are two main reasons that goods have to be sold by sequential auctions. First, units may not be available at the same time. They have to be sold separately. Second, committing to a future pricing policy is often impossible or illegal (Caillaud and Mezzetti, 2004). But for some goods, units may be available at the same time, for example, two bottles of French wine. Under this situation, the items can be sold either by sequential auctions or by simultaneous auctions.

Suppose that the two items in our model can be sold simultaneously. We discuss three simultaneous auctions: the discriminatory auction, the uniform-price auction, and the Vickrey auction. The equilibrium concept for the simultaneous auctions is the Bayesian Nash equilibrium (henceforth, BNE).⁷

⁷ For details of the Bayesian Nash equilibrium, please refer to Fudenberg and Tirole (1991).

For the discriminatory auction, the high-type bidder implements a mixed strategy and submits bids (b_1, b_2) with $b_1 \geq b_2$. Suppose that the highest bid in the second item is $\bar{b}$. Then it makes no sense for a player to bid more than $\bar{b}$ on the first unit, since any bid on the first unit that is greater than $\bar{b}$ will win with probability 1, and the bidder can do better by reducing it slightly. For the second item, she can win with probability $\underline{p}$ by bidding just above 0. Therefore, she submits b_2 such that:

$$\underline{p}\alpha\theta = (\underline{p} + \bar{p}G_1(b_2))(\alpha\theta - b_2),$$

where $G_1(b_2)$ is the probability that the high-type player's bid for the second unit exceeds her high-type opponent's bid for the first unit. If she bids $\bar{b}$, the above equation becomes $\underline{p}\alpha\theta = \alpha\theta - \bar{b}$. Thus, we can obtain $\bar{b} = \alpha\bar{p}\theta$. For the first unit the player bids b_1 so that:

$$\theta - \alpha\bar{p}\theta = (\underline{p} + \bar{p}G_2(b_1))(\theta - b_1),$$

where $G_2(b_1)$ is the probability that a high-type's first bid exceeds her high-type opponent's second. Hence, a high-type bidder will submit bids $b_1, b_2 \in [0, \alpha\bar{p}\theta]$ with c.d.f. $G_1(b_1) = \frac{pb_1}{\bar{p}(\alpha\theta - b_1)}$ and $G_2(b_2) = \frac{\bar{p}\theta(1-\alpha)+pb_2}{\bar{p}(\theta - b_2)}$, respectively. And her expected payoff is $\theta(1 + \alpha(\underline{p} - \bar{p}))$.

For the uniform-price auction, let the uniform price be the highest losing bid and denote it P . In such an auction, it is the weakly dominant strategy for the high-type to bid true value θ for the first item. To see why, first, she will not bid more than θ . Suppose she bids $b_1 < \theta$. If $\theta > P > b_1$, the bidder does not win any object. But if she were to raise b_1 to θ , then she may win a unit at a profitable price P . If $\theta > b_1 > P$, then raising her bid makes no difference. Thus, the bidder will bid her true value for the first item. For the second item, the high-type will bid just above 0. By doing so, the high-type's expected payoff for the uniform-price auction is $\theta(1 + \underline{p}\alpha)$.

In the Vickrey auction, the high-type bidder will bid her true values $(\theta, \alpha\theta)$. With probability $\underline{p}$, she faces a low-type opponent, and her payoff is $\underline{p}(\theta + \alpha\theta)$; with probability $\bar{p}$, she encounters another high-type bidder, and her payoff is $\bar{p}(\theta - \alpha\theta)$. Therefore, the total expected payoff for the high-type in the Vickrey auction is $\theta(1 + \alpha(\underline{p} - \bar{p}))$.

In summary of the above results, we have the following lemma:

Lemma 2. *In the discriminatory auction, the high-type buyer bids $b_1, b_2 \in [0, \alpha\bar{p}\theta]$ with c.d.f. $G_1(b_1) = \frac{pb_1}{\bar{p}(\alpha\theta - b_1)}$ and $G_2(b_2) = \frac{\bar{p}\theta(1-\alpha)+pb_2}{\bar{p}(\theta - b_2)}$, respectively, and her expected payoff is $\theta(1 + \alpha(\underline{p} - \bar{p}))$. In the uniform-price auction, the high-type bids true value θ for the first unit and bids just above 0 for the second unit, and her expected payoff is $\theta(1 + \underline{p}\alpha)$. In the Vickrey auction, the high-type bids her true values $(\theta, \alpha\theta)$, and her expected payoff is $\theta(1 + \alpha(\underline{p} - \bar{p}))$.*

Among the three simultaneous auctions with multi-unit demand, only the Vickrey auction is generally efficient, and the revenue equivalence principle does not hold anymore. In specific instances, however, there might be exceptions.⁸ Our model provides such a special case, in which the uniform-price auction is efficient and the discriminatory auction generates the same revenue as the Vickrey auction.

The uniform-price auction is efficient here due to the specific setting of the model: two items, two bidders with two types. If there is only one high-type bidder, she will win both item; if two high-type bidders compete, each wins one item. Meanwhile, the

uniform-price auctions generate very low revenue for the seller due to the extreme “demand-reduction effect”: The sale price is either 0 when there is at least one low-type bidder or infinitesimally more than 0 if there are no low-type bidders. The discriminatory auction is not efficient here, but it generates the same revenue as the Vickrey auction. We believe that it is a special case due to our specific setting.

Now we are ready to compare the revenues for the seller for all of the auction formats discussed above. It is natural that the sequential SPA and the Vickrey auction generate the same revenue. Due to the specific setting of our model, the discriminatory auction generates the same revenue as the Vickrey auction and the sequential SPA. As explained above, the uniform-price auction generates the lowest revenue. And compared to the sequential SPA, the seller obtains less in the sequential FPA. Therefore, we have the following proposition:

Proposition 2. *In the multi-unit demand auctions, the discriminatory auction, the Vickrey auction, and the sequential SPA generate the same revenue for the seller, followed in order by the sequential FPA, and by the uniform-price auction.*

4. Conclusion

Most of the sequential auction literature is restricted to single-unit demand. Within the IPV paradigm, this note discusses two-round sequential auctions with multi-unit demands. We first characterize the equilibrium for the sequential FPA, and show that the expected price in the second round is strictly higher than that of the first. Then we compare the revenues for the sequential FPA, the sequential SPA, and the simultaneous auctions. We show that the discriminatory auction, the Vickrey auction, and the sequential SPA generate the same revenue for the seller, followed in order by the sequential FPA, and by the uniform-price auction.

This note serves as one of the first steps to analyze the sequential FPA with multi-unit demand, and compare its revenue with other sequential and simultaneous auctions. However, there are limitations of this research. For example, the two-item, two-bidder, and two-type environment of our model is very specific; one of the key results on the price path is derived only numerically since we are not able to obtain a closed form solution. It would be interesting to extend our model to more general settings. First, it could be generalized to N bidders with N types or continuous types in N -round sequential auctions. Second, we could introduce risk averse bidders to the model. Third, we could introduce reserve prices to the sequential auctions. These interesting extensions are challenging but deserve further investigation.

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References

- Caillaud, B., Mezzetti, C., 2004. Equilibrium reserve prices in sequential ascending auctions. *Journal of Economic Theory* 117, 78–95.
- Engelbrecht-Wiggans, R., 1994. Sequential auctions of stochastically equivalent objects. *Economics Letters* 44, 87–90.
- Enkhbayar, T., 2004. Information Transmission in Repeated Auctions. Working Paper, University of Tokyo.

⁸ Krishna (2002, p. 202) shows an example in which all three auctions are efficient and the revenue equivalence principle holds.

- Fevrier, P., 2003. He who must not be named. *Review of Economic Design* 8, 99–119.
- Fudenberg, D., Tirole, J., 1991. *Game Theory*. The MIT Press.
- Hausch, D., 1986. Multi-object auctions: sequential vs. simultaneous sales. *Management Science* 32, 1599–1610.
- Hausch, D., 1988. A model of sequential auctions. *Economics Letters* 26, 227–233.
- Jeitschko, T.D., 1998. Learning in sequential auctions. *Southern Economic Journal* 65, 98–112.
- Katzman, B., 1999. A two stage sequential auctions with multi-unit demands. *Journal of Economic Theory* 86, 77–99.
- Krishna, V., 2002. *Auction Theory*. Academic Press.
- Liu, Q., 2011. Equilibrium of a sequence of auctions when bidders demand multiple items. *Economics Letters* 112, 192–194.
- Maskin, E., Riley, J., 1985. Auction theory with private values. *American Economic Review* 75 (2), 150–155.
- Maskin, E., Riley, J., 1989. Optimal multi-unit auctions. In: Hahn, F. (Ed.), *The Economics of Missing Markets*. pp. 312–335.
- McAfee, P., Vincent, D., 1993. The declining price anomaly. *Journal of Economic Theory* 60, 191–212.
- McAfee, P., Vincent, D., 1997. Sequentially optimal auctions. *Games and Economic Behavior* 18, 246–276.
- Milgrom, P., Weber, R., 2000. A theory of auctions and competitive bidding II. In: Klemperer, P. (Ed.), *The Economic Theory of Auctions*, Vol. 2. pp. 179–194.
- Myerson, R., 1981. Optimal auction design. *Mathematics of Operation Research* 6, 58–73.
- Ortega-Reichert, A., 1968. *Models for Competitive Bidding Under Uncertainty*. Ph.D. Thesis and Technical Report No. 8, Department of Operations Research, Stanford University (Chapter VIII).
- Riley, J., Samuelson, W., 1981. Optimal auctions. *American Economic Review* 71, 381–392.
- Thomas, C., 2010. Information revelation and buyer profits in repeated procurement competition. *The Journal of Industrial Economics* 58, 79–105.
- Vickrey, W., 1961. Counterspeculation, auctions, and competitive sealed tenders. *Journal of Finance* 16, 8–37.
- Weber, R., 1983. Multiple objects auctions. In: Engelbrecht-Wiggans, R., Shubik, M., Stark, R. (Eds.), *Auctions, Bidding and Contracting: Uses and Theory*. New York University Press.